(ii) Regression coefficient *of Y* on *X*:

$$b_{yx} = \frac{N\Sigma XY-(\Sigma X)(\Sigma Y)}{N\,\Sigma X^2-(\Sigma X)^2}$$

$$= \frac{7(334)-(28)(77)}{7(140)-(28)^2}$$

$$= \frac{2338-2156}{980-784}$$

$$= \frac{182}{196}$$

$$\therefore b_{yx} = 0.929$$

(iii) Regression equation of *Y* on *X*:

$$Y-\overline{Y} = b_{yx}(X-\overline{X})$$

$$Y-11 = 0.929\,(X-4)$$

$$Y = 0.929X-3.716+11$$

$$Y = 0.929X+7.284$$

$\therefore$ The regression equation of Y on X is

$$Y = 0.929X + 7.284$$

Example 9.10

Calculate the two regression equations of X on Y and Y on X from the data given below, taking deviations from a actual means of X and Y.

Price (₹)	10	12	13	12	16	15
Amount demanded	40	38	43	45	37	43

Estimate the likely demand when the price is ₹ 20.

Solution:

Calculation of Regression equation

(i) Regression equation of *X* on *Y*:

$$X-\overline{X} = r\,\frac{\sigma_x}{\sigma_y}(Y-\overline{Y})$$

From the Table 9.8, we get

$$\overline{X} = \frac{78}{6}=13\,,\ \overline{Y} = \frac{246}{6}=41$$

$$b_{xy} = r\,\frac{\sigma_x}{\sigma_y} = \frac{\Sigma xy}{\Sigma y^2} = \frac{-6}{50}=-0.12$$

$$X-13 = -0.12\,(Y-41)$$

$$X-13 = -0.12Y+4.92$$

$$X = -0.12Y+17.92$$

X	$x = (X-13)$	x^2
10	–3	9
12	–1	1
13	0	0
12	–1	1
16	3	9
15	2	4
$\Sigma X = 78$	$\Sigma x = 0$	$\Sigma x^2 = 24$

Y	$y = (Y-41)$	y^2	xy
40	–1	1	3
38	–3	9	3
43	2	4	0
45	4	16	–4
37	–4	16	–12
43	2	4	4
$\Sigma Y = 246$	$\Sigma y = 0$	$\Sigma y^2 = 50$	$\Sigma xy = -6$

Table 9.8

(ii) Regression Equation of Y on X:

$$Y-\overline{Y} = r\,\frac{\sigma_y}{\sigma_x}(X-\overline{X})$$

$$b_{yx} = r\,\frac{\sigma_y}{\sigma_x} = \frac{\Sigma xy}{\Sigma x^2} = -\frac{6}{24}=-0.25$$

$$Y-41 = -0.25\,(X-13)$$

$$Y-41 = -0.25\,X+3.25$$

$$Y = -0.25\,X+44.25$$

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